

Nepal Flood Risk Prediction & Early Warning System

AI-Powered Flood Intelligence Platform using Machine Learning

Dataset	Nepal Flood & Weather 2023–2026 (Kaggle)
Problem Type	Multi-class Classification (3 classes)
Best Model	Random Forest Classifier
Weighted F1 Score	87.65%
ROC-AUC	0.9788
Training Records	10,712
Test Records	2,678
Total Features	31 (14 base + 17 engineered)
Platform	Python · scikit-learn · Streamlit · Plotly

Table of Contents

1. Project Overview & Objectives
2. Dataset Description
3. Exploratory Data Analysis
4. Data Preprocessing Pipeline
5. Feature Engineering
6. Machine Learning Pipeline
7. Model Evaluation & Results
8. Feature Importance & Explainability
9. System Architecture
10. Streamlit Application
11. Key Findings & Insights
12. Limitations & Future Work
13. Technologies Used

1. Project Overview & Objectives

The **Nepal Flood Risk Prediction & Early Warning System** is an end-to-end Machine Learning application that analyses daily weather and hydrological observations from Nepal's major river monitoring stations to predict flood risk levels. The system is designed as a **National Flood Intelligence Center**, combining data analytics, predictive modelling, and a professional interactive web dashboard.

Objectives

- Predict flood risk level (LOW / MEDIUM / HIGH) from observable weather and environmental parameters
- Identify the most critical variables driving flood risk across Nepal's river basins
- Deliver an accessible, professional early-warning interface for monitoring stations
- Demonstrate a complete, reproducible ML pipeline from raw data to deployed prediction

Problem Type: Multi-class classification (3 classes: LOW, MEDIUM, HIGH risk)
Approach: Supervised learning with engineered features and SMOTE-balanced training
Evaluation Strategy: Time-aware train/test split to simulate real-world forecasting

2. Dataset Description

The dataset is sourced from Kaggle: *Nepal Flood & Weather Dataset 2023–2026* by Tejal Kunjir. It contains daily hydrological and meteorological records from 10 monitoring stations across 9 major river basins in Nepal, covering January 2023 to August 2026.

Total Records	Date Range	Stations	Rivers	Missing Values	Duplicates
13,390	Jan 2023 – Aug 2026	10	9	0	0

Raw Feature List

Feature	Type	Description
date	DateTime	Observation date (daily)
location	Categorical	Monitoring station name (10 unique)
river	Categorical	River name (9 unique)
basin	Categorical	River basin (9 unique)
latitude / longitude	Float	Station geographic coordinates
elevation_m	Integer	Station elevation in metres
precipitation_mm	Float	Daily total precipitation
rain_mm	Float	Daily rainfall accumulation
soil_moisture_0_100cm_m3m3	Float	Volumetric soil moisture (0–100 cm)
temperature_mean_c	Float	Daily mean temperature
dew_point_mean_c	Float	Daily mean dew point temperature
precipitation_hours	Integer	Hours with measurable precipitation

Feature	Type	Description
relative_humidity_mean_pct	Integer	Daily mean relative humidity (%)
wind_speed_max_kmh	Float	Maximum daily wind speed
wind_gusts_max_kmh	Float	Maximum daily wind gusts
wind_direction_dominant_deg	Integer	Dominant wind direction (degrees)
river_discharge_m3s	Float	River discharge (m³/s) — used to derive target

Monitoring Stations & Discharge Thresholds

Station	River	p75 (m³/s)	p90 (m³/s)
Kusum	West Rapti	176.22	415.87
Belsot	Kamala	43.82	91.77
Chameliya/Nayalbadi	Chamelia	43.06	87.39
Bahrabise	Bhote Koshi	9.52	14.01
Devghat	Narayani	3.56	6.04
Chatara	Saptakoshi	2.53	4.33
Khokana	Bagmati	2.31	4.94
Rasuwagadhi	Bhote Koshi	2.63	3.66
Chisapani	Karnali	1.40	2.38
Bhada Bridge	Babai	0.33	2.29

Note: Per-station thresholds are critical. Kusum's p90 is 415.87 m³/s while Bhada Bridge's is only 2.29 m³/s — a global threshold would be meaningless.

3. Exploratory Data Analysis

Target Variable Distribution

Class	Label	Condition	Count	Share
0	LOW	discharge < station p75	10,032	74.9%
1	MEDIUM	p75 ≤ discharge < p90	2,013	15.0%
2	HIGH	discharge ≥ p90	1,345	10.0%

The dataset is imbalanced (75% LOW, 15% MEDIUM, 10% HIGH). SMOTE oversampling was applied to the training set to address this.

Precipitation Analysis

Statistic	Value
Daily Mean Precipitation	4.20 mm

0.09 mm/day average

River Discharge Analysis

3,175.87 (Kusum station)

Seasonal HIGH-Risk Concentration

0.0%

Feature–Risk Correlations (absolute Pearson r)

0.150

4. Data Preprocessing Pipeline

The preprocessing pipeline is implemented in `src/data_preprocessing.py` and executes the following reproducible steps:

- **1. Duplicate removal** — no duplicates found; step retained for robustness
- **2. Date parsing** — date column parsed to datetime64 format
- **3. Sorting** — records sorted by location + date for time-series consistency
- **4. Date feature extraction** — year, month, day_of_year, ISO week
- **5. Cyclical month encoding** — month_sin and month_cos to capture seasonality without ordinal artifacts
- **6. Missing value imputation** — median fill per column (no missing values present; kept for robustness)
- **7. Outlier capping** — 1st–99th percentile winsorisation on precipitation, rainfall, and wind features to preserve real flood extremes
- **8. Target variable derivation** — per-station discharge percentile thresholds (p75 = MEDIUM boundary, p90 = HIGH boundary)

Data Leakage Prevention: Flood risk thresholds (p75, p90) are computed on the full dataset and saved to disk. In a production setting these would be computed only on training folds. River discharge is excluded from the model feature set as it is the source of the target label.

5. Feature Engineering

Implemented in `src/feature_engineering.py`. Starting from 14 base features, the pipeline engineers 17 additional features for a total of **31 model features**.

Rolling Window Features

Window	Variables	Aggregations	Features Added
3-day	precipitation, rain, humidity	mean, sum	6
7-day	precipitation, rain, humidity	mean, sum	6
7-day	precipitation only	max	1

Leakage prevention: All rolling features use `shift(1)` before rolling, ensuring only past values (not the current day) are used.

Domain-Engineered Features

Feature	Formula	Rationale
saturation_index	$\text{soil_moisture} \times (\text{humidity} / 100)$	Catchment saturation proxy
heavy_rain_flag	1 if precipitation > 20 mm	Binary heavy-rain event signal
temp_dew_spread	temperature – dew_point	Low spread = near-saturated atmosphere
month_sin / cos	$\sin / \cos \text{ of } 2\pi \times \text{month} / 12$	Cyclical seasonal encoding
location_enc	LabelEncoder	Station identity for cross-station differentiation

6. Machine Learning Pipeline

Train / Test Split

Split	Records	Date Range (approx.)
Training set	10,712 (80%)	Jan 2023 – mid 2025
Test set	2,678 (20%)	mid 2025 – Aug 2026

A **time-aware split** is used: the last 20% of chronologically sorted records form the test set. This simulates real-world forecasting where the model must generalise to future, unseen dates — not random held-out samples.

Class Imbalance — SMOTE

SMOTE (Synthetic Minority Oversampling Technique) is applied **only to the training set**. It generates synthetic samples for MEDIUM and HIGH classes to achieve a balanced class distribution of 7,830 samples per class after resampling. The test set is kept unmodified to reflect real-world class proportions.

Models Evaluated

Model	Key Hyperparameters
Logistic Regression	max_iter=1000, class_weight=balanced
Decision Tree	max_depth=10, class_weight=balanced
Random Forest	n_estimators=200, max_depth=15, class_weight=balanced, n_jobs=-1
Gradient Boosting	n_estimators=200, max_depth=5, learning_rate=0.1

7. Model Evaluation & Results

Model	Accuracy	Precision	Recall	F1 (weighted)	ROC-AUC
Logistic Regression	83.83%	88.93%	83.83%	85.55%	0.9613
Decision Tree	84.13%	88.14%	84.13%	85.77%	0.9542
Random Forest ★	86.71%	89.17%	86.71%	87.65%	0.9788
Gradient Boosting	86.22%	88.91%	86.22%	87.26%	0.9751

★ **Selected model: Random Forest** — highest weighted F1-score (87.65%) and ROC-AUC (0.9788). Weighted F1 is the primary metric because the test set is imbalanced.

Classification Report — Random Forest (Test Set)

Class	Precision	Recall	F1-Score	Support
LOW	0.99	0.94	0.97	2,202
MEDIUM	0.47	0.44	0.46	294
HIGH	0.38	0.64	0.48	182
Weighted Avg	0.89	0.87	0.88	2,678

LOW class is predicted with very high accuracy (F1=0.97). MEDIUM and HIGH classes are harder to classify due to their smaller representation and overlapping feature distributions. HIGH recall of 0.64 indicates the model successfully flags most genuine flood-risk events, which is operationally the most important outcome.

8. Feature Importance & Explainability

Random Forest feature importances are computed via mean decrease in Gini impurity across all 200 trees. The top 15 features are shown below (out of 31 total):

Feature	Importance	%
Month (sin)		11.47%
Day of Year		9.70%
Dew Point (°C)		8.31%
Humidity 7-day Avg		5.86%
Humidity 7-day Sum		5.54%
Rain 7-day Total		4.31%
Rain 7-day Avg		4.18%
Temperature (°C)		3.84%
Soil Moisture		3.76%
Humidity 3-day Sum		3.74%
Month (cos)		3.68%
Precip 7-day Total		3.44%
Saturation Index		3.33%
Humidity 3-day Avg		3.25%
Precip 7-day Avg		3.09%

Key insight: Seasonality features (month_sin, day_of_year) dominate because flood risk in Nepal is overwhelmingly monsoon-driven (93.3% of HIGH events in Jun–Sep). After seasonality, dew point, rolling humidity windows, and rolling rainfall are the strongest predictors — reflecting cumulative moisture build-up that precedes flooding.

Feature importance reflects how strongly variables contribute to model predictions across the training dataset. It does not establish direct causation.

9. System Architecture

Component	File	Responsibility
Data Ingestion	data/CORRECTED_...csv	Raw daily weather & hydrological records

Component	File	Responsibility
Preprocessing	src/data_preprocessing.py	Cleaning, label derivation, date features
Feature Engineering	src/feature_engineering.py	Rolling windows, encoding, domain features
Model Training	src/train_model.py	SMOTE, fit 4 classifiers, evaluate, save
Inference	src/predict.py	Live prediction from user input
Utilities	src/utlis.py	Shared helpers, metrics loading, formatting
Model Artifacts	models/	*.pkl, thresholds.csv, metrics.json, plots
Dashboard	app.py	Streamlit 7-page professional dashboard
Notebook	notebooks/EDA_and_Modeling.ipynb	53-cell EDA and modelling walkthrough

10. Streamlit Application

The application is built as a **National Flood Intelligence Center** with a professional dark-navy sidebar navigation and 7 functional pages:

Page	Description
Dashboard	Command center: KPI cards, live risk status, donut chart, seasonal pattern, precipitation trend, station network
Flood Risk Prediction	Grouped input form (weather / environmental conditions), real-time model inference, prediction card with color-coded risk
Data Analysis	4-tab interactive workspace: Weather Analysis, Hydrological Analysis, Risk Analysis, Correlations — with interactive maps
Model Performance	Metric cards, model comparison chart (F1 + AUC), confusion matrix, feature importance chart, model summary
Station Monitoring	Per-station deep-dive: KPIs, discharge timeline, precipitation timeline, monthly risk breakdown
Risk Trends	Early Warning Center: alert cards per station, 30-day rolling risk trend lines, scrollable risk event timeline
Regional Insights	Basin-level risk profiles, elevation vs risk scatter, season × basin heatmap

11. Key Findings & Insights

Extreme Seasonal Concentration

93.3% of all HIGH-risk events occur during the monsoon season (June–September). Risk is negligible in winter, confirming the dominant role of the South Asian monsoon in Nepal's flood cycle.

Peak Risk Month: July

July records the highest average daily precipitation (12.55 mm) and the highest frequency of HIGH and MEDIUM risk events. Monthly discharge is 10× higher in July–August compared to winter months.

Cumulative Moisture Matters Most

Rolling 7-day humidity and rainfall windows rank among the top 5 features. This confirms that antecedent soil saturation and atmospheric moisture are stronger predictors than single-day precipitation alone.

Dew Point as a Flood Predictor

Dew point has the highest single-feature correlation with flood risk ($r=0.545$). A high dew point indicates a saturated, moisture-laden atmosphere — conditions that reliably precede heavy rainfall and elevated discharge.

Per-Station Thresholds Are Essential

Kusum's p90 threshold is 415.87 m³/s while Bhada Bridge's is only 2.29 m³/s. A global threshold would misclassify nearly all records at smaller stations, making per-station normalisation a critical design decision.

Random Forest Outperforms All Baselines

Random Forest achieves 87.65% weighted F1 and 0.9788 ROC-AUC — outperforming Gradient Boosting (87.26%), Decision Tree (85.77%), and Logistic Regression (85.55%). Its ensemble structure handles the non-linear, strongly seasonal patterns effectively.

12. Limitations & Future Work

Current Limitations

- **No real-time data feed** — predictions require manually entered parameters; no live API integration
- **MEDIUM/HIGH class performance** — F1 scores of 0.46–0.48 for minority classes; additional training data would help
- **No spatial modelling** — geographic proximity between stations is not exploited; upstream conditions do not inform downstream predictions
- **Static thresholds** — p75/p90 thresholds computed on the full dataset; production implementation should use training-only data per cross-validation fold
- **No snowmelt features** — for high-elevation stations (e.g., Rasuwagadhi at 1,749 m), snowmelt is a significant flood driver not captured in the current feature set

Future Improvements

- Integrate OpenWeatherMap or DHM Nepal real-time API for live data ingestion
- Implement XGBoost / LightGBM with Bayesian hyperparameter optimisation
- Add LSTM or Temporal Fusion Transformer for sequence-aware multi-step forecasting
- Build upstream-to-downstream station routing to capture catchment-level signals
- Add SHAP values for per-prediction explainability beyond global feature importance
- Deploy on Streamlit Community Cloud or AWS EC2 for public access

13. Technologies Used

Category	Technology	Purpose
Language	Python 3.10+	Complete backend and frontend
Data	pandas, numpy	Data manipulation and numerical computation
ML Framework	scikit-learn 1.4+	Model training, evaluation, preprocessing
Imbalance	imbalanced-learn	SMOTE oversampling
Visualisation	Plotly, matplotlib, seaborn	Interactive charts and saved EDA plots
Frontend	Streamlit 1.35+	7-page web dashboard

Category	Technology	Purpose
Persistence	joblib	Model and encoder serialisation
Notebooks	Jupyter / nbformat	53-cell EDA and modelling notebook
Dataset Source	Kaggle	Nepal Flood & Weather Dataset 2023–2026

Nepal Flood Risk Prediction & Early Warning System

All metrics, statistics, and model results in this report are derived from the actual trained Random Forest classifier and the Nepal Flood & Weather Dataset 2023–2026.

AI/ML · Weather Intelligence · Flood Risk Analytics · Python · scikit-learn · Streamlit